

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CG

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INDEX NUMBER

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BIOLOGY

9744/02

Paper 2 Structured Questions

28 August 2018

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, CG and index number on page 1, 11 and 19 in the spaces provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	18
2	10
3	10
4	10
5	10
6	11
7	10
8	8
9	7
10	6
Total	100

This document consists of **27** printed pages and **1** blank page.



Answer **all** questions.

- 1 Fig. 1.1 shows an electron micrograph of a pancreatic cell that secretes large amounts of insulin that helps to regulate blood glucose level.

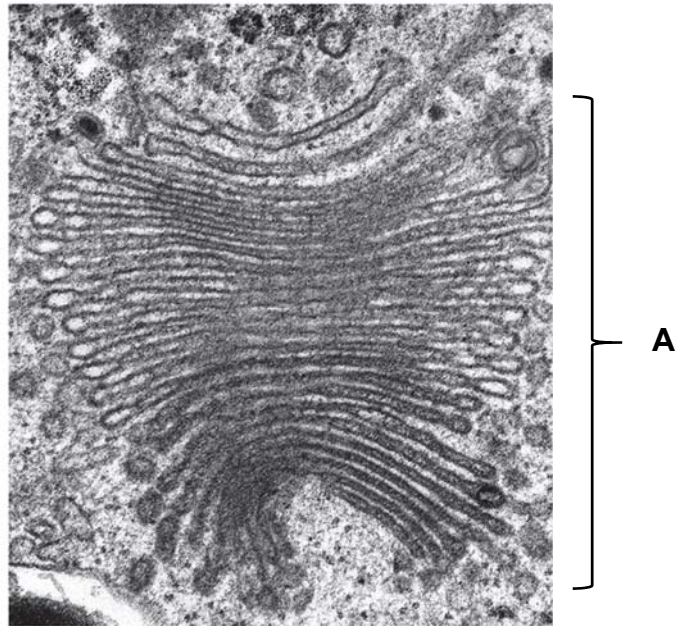


Fig. 1.1

(a) With reference to Fig. 1.1,

(i) identify organelle **A**;

..... [1]

(ii) describe **two** identifying features of organelle **A** that allows its identification in (a)(i).

.....

 [2]

Fig. 1.2 shows a diagram of the molecular structures of tristearin (a triglyceride) and phosphatidylcholine (a phospholipid).

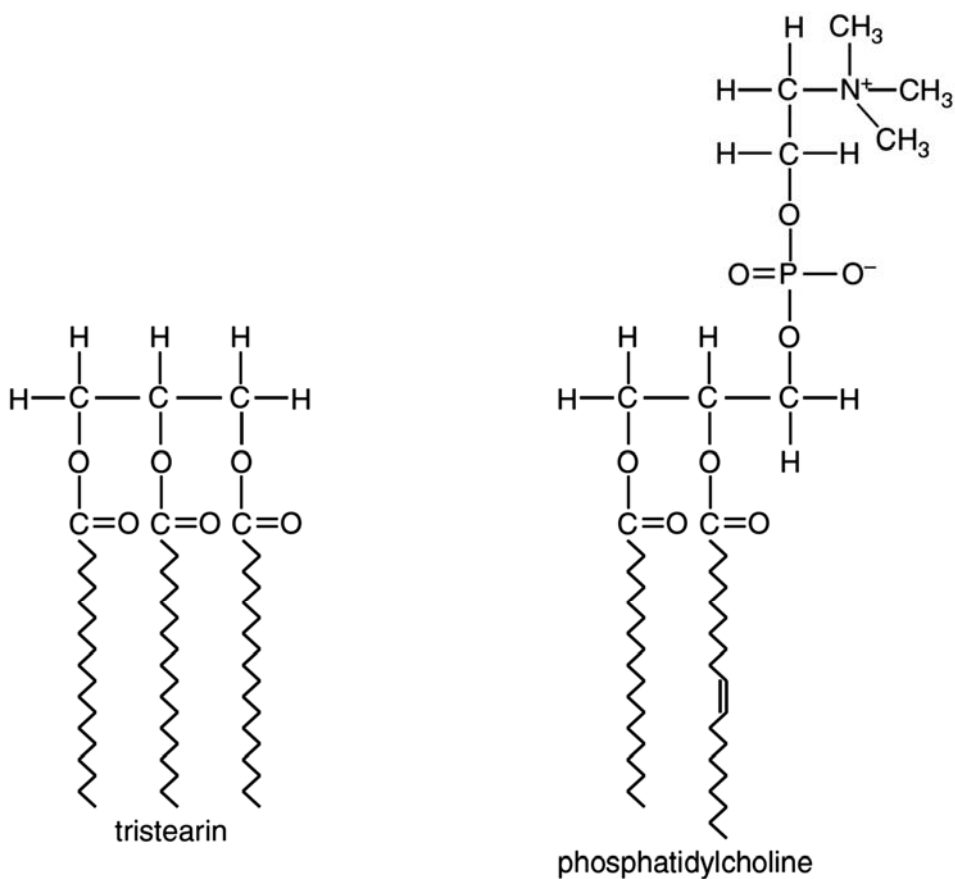


Fig. 1.2

(b) Table 1 shows a structural difference between the two molecules shown in Fig. 1.2.

Complete Table 1 with two further **structural** differences **other than** in numbers of different types of atoms.

Table 1

structural feature	tristearin	phosphatidylcholine
length of fatty acid chain	all same length	different lengths

[2]

- (c) Triglyceride is used as energy storage while phospholipids are membrane components. Explain why phospholipids are suitable membrane components but not triglyceride.

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.....

.....

..... [2]

Cells in the pancreas secrete enzymes, such as amylase and trypsin, into a duct. The enzymes are packaged in vesicles so that they can be exported from these cells as shown in Fig. 1.3.

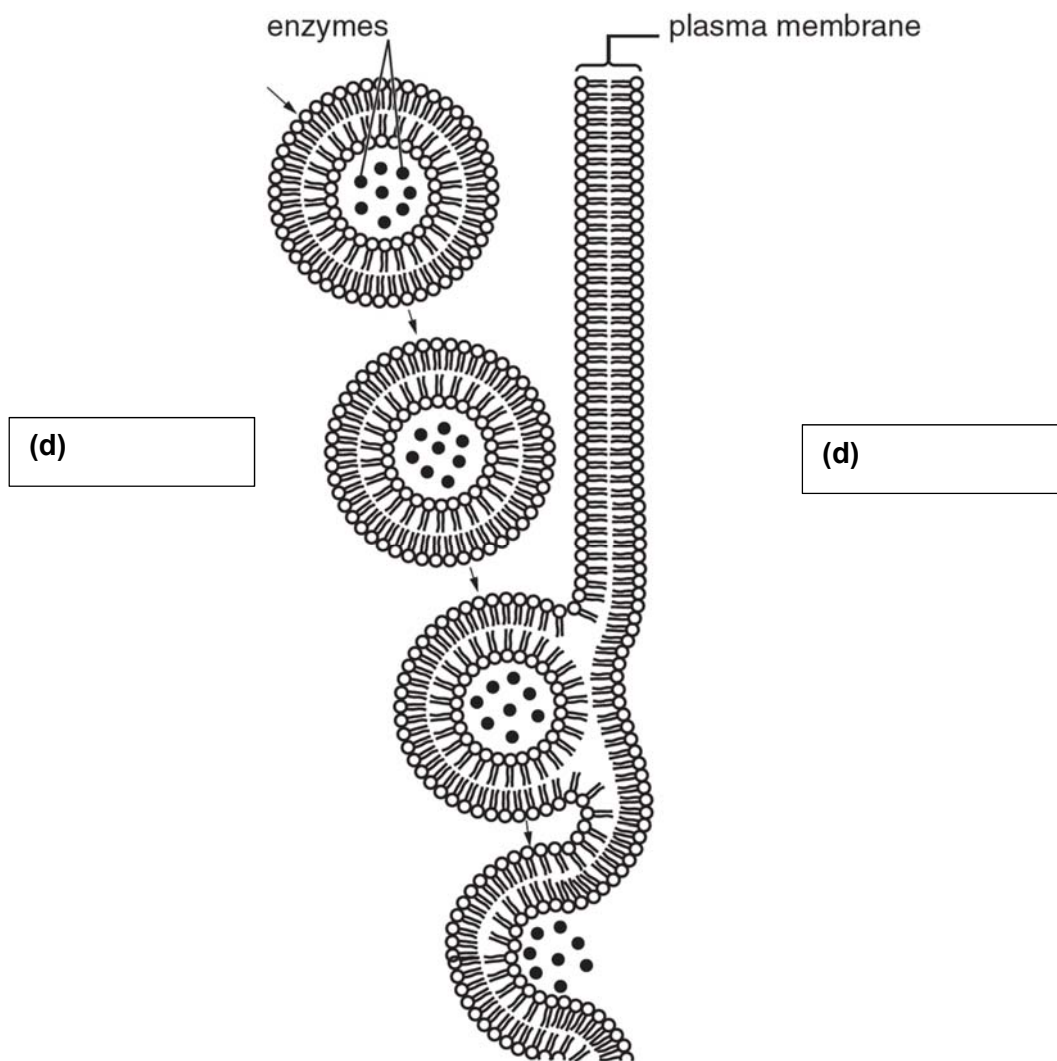


Fig. 1.3

- (d) On Fig. 1.3, label the cytoplasm of the cell as '**cytoplasm**' and extracellular fluid as '**extracellular**'. [1]

- (e)** With reference to Fig. 1.3,

- (i) explain how enzymes that are secreted by cells in the pancreas are packaged into vesicles and exported, after their synthesis at the endoplasmic reticulum.

This image shows a blank sheet of primary-ruled paper. It features ten sets of horizontal lines across its width. Each set consists of three lines: two dashed lines forming the main body of the writing area and one solid line at the top or bottom of each set. The lines are evenly spaced and extend across the entire width of the page.

[6]

- (ii) explain **one** property of the plasma membrane that allows vesicle formation.

[2]

- (f) Describe **two** advantages of having plasma membranes **within** the cell.

[2]

[Total: 18]

2 Fig. 2.1 shows a bacterium cell.

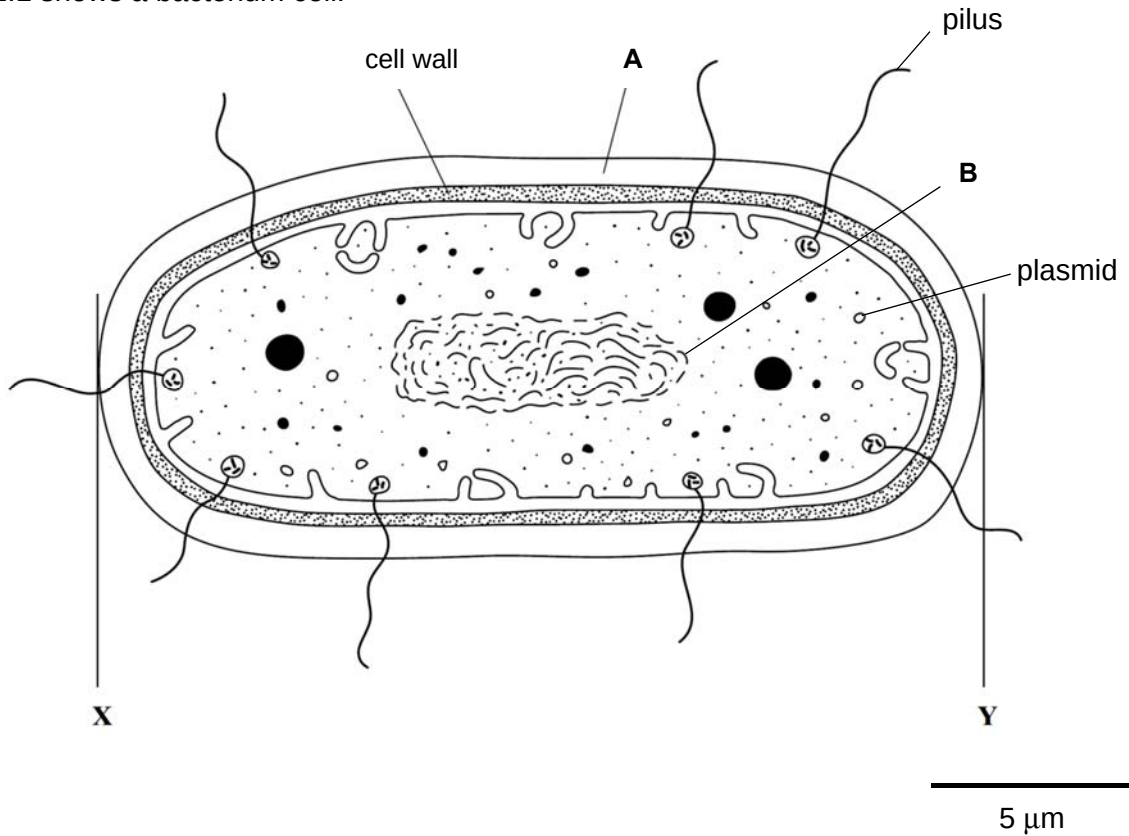


Fig. 2.1

(a) Identify structures **A** and **B**.

A

B

[2]

(b) Calculate the length of the bacterium cell from **X** to **Y** in μm . Show your working clearly.

[1]

Bacterial cells can acquire new alleles through genetic transfers between bacterial cells. One way is through plasmids called F plasmids in bacterial cells, another is through bacteriophages.

(c) Describe how a bacterium cell can acquire new alleles through F plasmids.

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.....

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.....

.....

..... [3]

An investigation was conducted on a temperate phage that infects a new strain of bacteria, to find the prophage insertion site on the bacterial chromosome.

Fig. 2.2 shows the chromosome map of the bacterial strain. The chromosome map shows positions of the genes *ade*, *arg*, *cys*, *his*, *leu* and *pro*, which are each involved the synthesis of amino acids essential for bacterial growth.

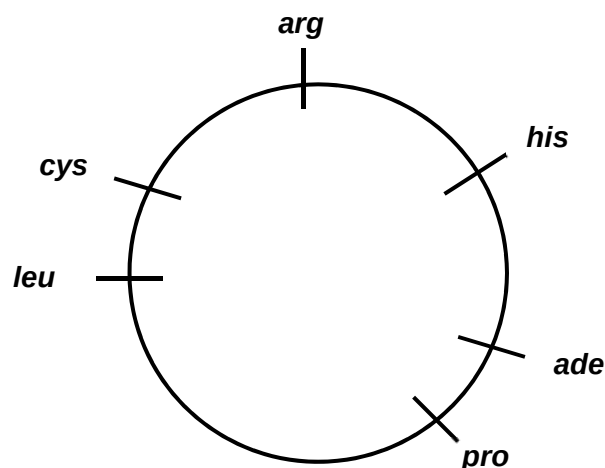


Fig. 2.2

The bacterial strain of genotype *ade*⁺ *arg*⁺ *cys*⁺ *his*⁺ *leu*⁺ *pro*⁺ (where '+' indicates functional wild-type allele) was used as a source of the phage. The phages were then added to a bacterial strain of genotype *ade*⁻ *arg*⁻ *cys*⁻ *his*⁻ *leu*⁻ *pro*⁻ (where '-' indicates non-functional allele).

After a short incubation, samples of these bacteria were plated on six different media. The plates were then observed for growth of bacterial colonies. Table 2 shows the results of the experiment.

Table 2

medium	nutrient supplementation in medium						presence of colonies
	ade	arg	cys	his	leu	pro	
1	✗	✓	✓	✓	✓	✓	N
2	✓	✗	✓	✓	✓	✓	N
3	✓	✓	✗	✓	✓	✓	C
4	✓	✓	✓	✗	✓	✓	N
5	✓	✓	✓	✓	✗	✓	C
6	✓	✓	✓	✓	✓	✗	N

Key:

‘✓’ indicates presence of amino acid

‘✗’ indicates absence of nutrient supplement

‘C’ indicates presence of colonies,

‘N’ indicates absence of colonies

(d) With reference to Table 2,

(i) indicate on Fig. 2.2, the position of prophage insertion. [1]

(ii) explain why colonies are able to grow on medium **3** and **5**.

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.....

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.....

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..... [3]

[Total: 10]

- 3 Fig. 3.1 shows the structure of a mature tRNA that is involved in the production of a globular enzyme in a eukaryotic cell.

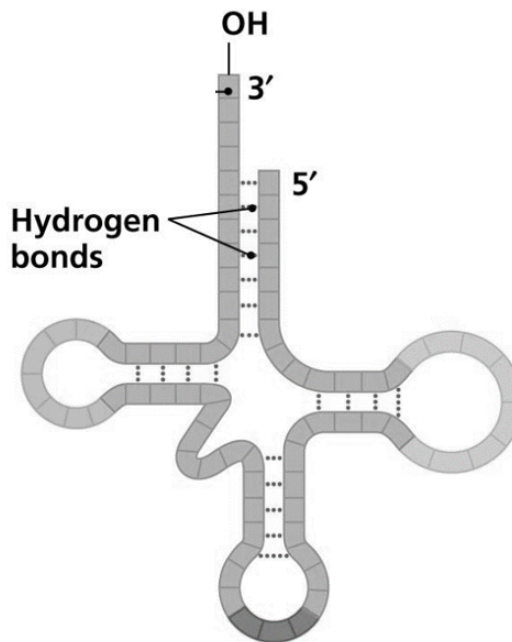


Fig. 3.1

- (a) With reference to Fig. 3.1, explain the roles of hydrogen bonds in the proper functioning of tRNA.

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.....

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.....

.....

..... [3]

- (b) The length of the tRNA gene is longer than that of the mature tRNA.

Outline how a **mature** tRNA molecule is synthesised in eukaryotes.

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..... [3]

Fig. 3.2 shows a pathway in the regulation of gene expression to synthesise enzymes in a eukaryotic cell can be achieved.

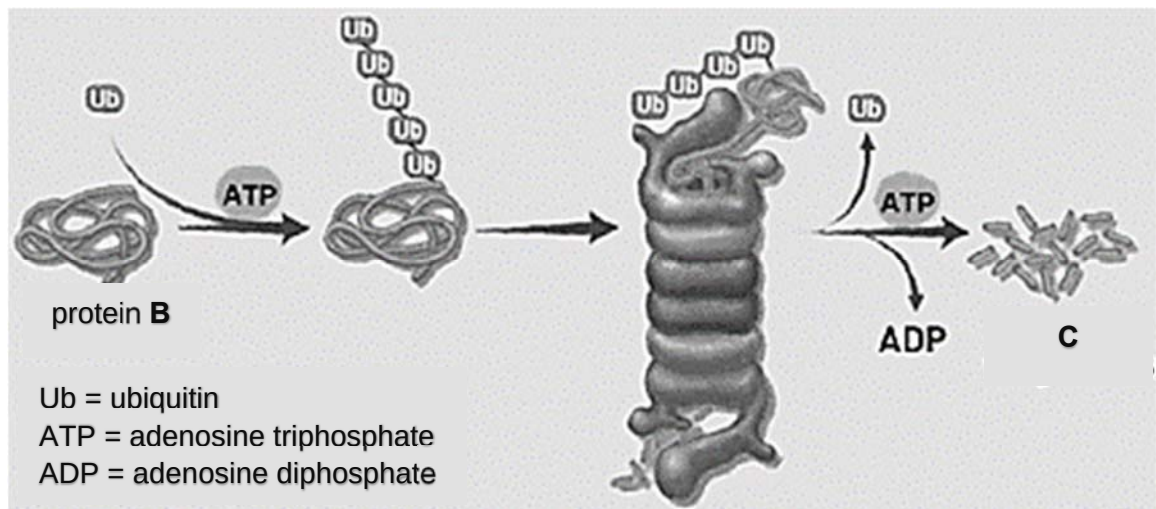


Fig. 3.2

(c) With reference to Fig. 3.2,

(i) name the process involved;

..... [1]

(ii) explain how product **C** is formed from protein **B**.

.....

 [3]

[Total: 10]

- 4 Crossing over between homologous chromosomes is an important process that gave rise to genetic variation.

Errors during crossing over can result in abnormal chromosomes that will affect the phenotype of an organism.

Fig. 4.1 show the result of an unequal crossing over event in prophase I.

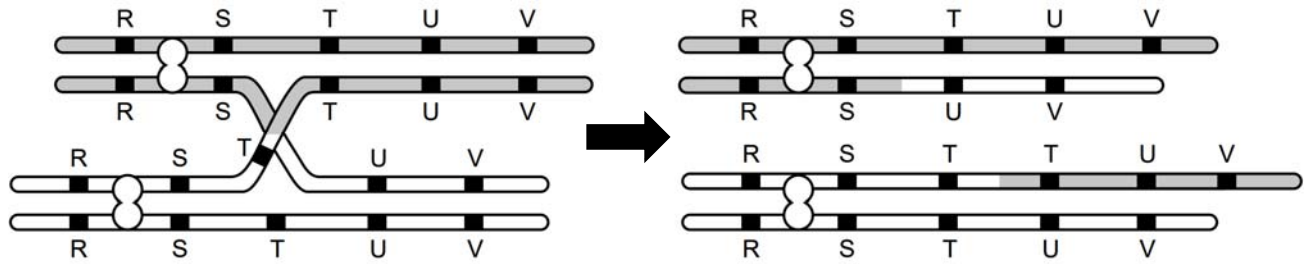


Fig. 4.1

- (a) (i) State the type of mutation that has occurred in Fig. 4.1.

[1]

- (ii) Suggest the possible effects on the phenotype of the organism if protein T is a hormone involved in metabolism.

[2]

Fig. 4.2 shows the human karyotype of one of the most common chromosomal abnormalities. The affected embryos or foetuses with this condition ended up in miscarriages in the first trimester.

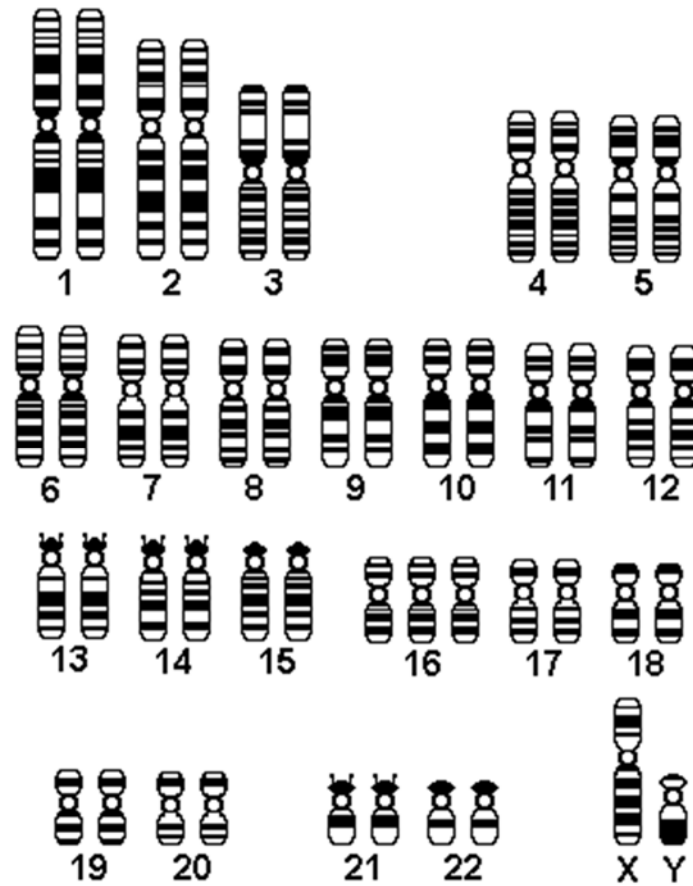


Fig. 4.2

(b) With reference to Fig. 4.2, explain how this condition arises.

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[4]

- (c) Fig. 4.1 and Fig 4.2 are examples of chromosome aberration.

Use examples other than the ones shown, outline what does *chromosome aberration* mean.

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.....

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..... [3]

[Total: 10]

- 5 Two independently assorting genes coding for eye colour and the presence of wing crossveins in the *Drosophila* fruit fly were investigated in a few genetic crosses with different adults.

The outcomes of four crosses are shown below in Table 5.

Table 5

Cross	Parental phenotypes	Orange, crossveins present	Orange, crossveins absent	Red, crossveins present	Red, crossveins absent
1	orange, crossveins present, orange, crossveins present	83	26	0	0
2	red, crossveins present, red, crossveins absent	20	18	65	63
3	red, crossveins absent, red, crossveins present	0	0	71	81
4	red, crossveins present, red, crossveins present	28	11	93	34

(a) Using suitable symbols,

- (i) deduce the genotypes of the adults in Cross 2.

Red, crossveins present

Red, crossveins absent [2]

- (ii) explain why Cross 1 and Cross 3 have zero offspring in some phenotypic categories.

.....

 [2]

(b) In order to test the significance of differences between observed and expected results, a statistical test was carried out for Cross 4.

- (i) Name the statistical test carried out.

..... [1]

- (ii) State the expected offspring ratio.

..... [1]

- (iii) Explain what does it mean when the results of the chi squared test has a probability of more than 0.05.

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..... [2]

- (c) Using a named example, describe how the environment may affect the phenotype of organisms.

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..... [2]

[Total: 10]

6 Fig. 6.1 shows a section of a chloroplast.

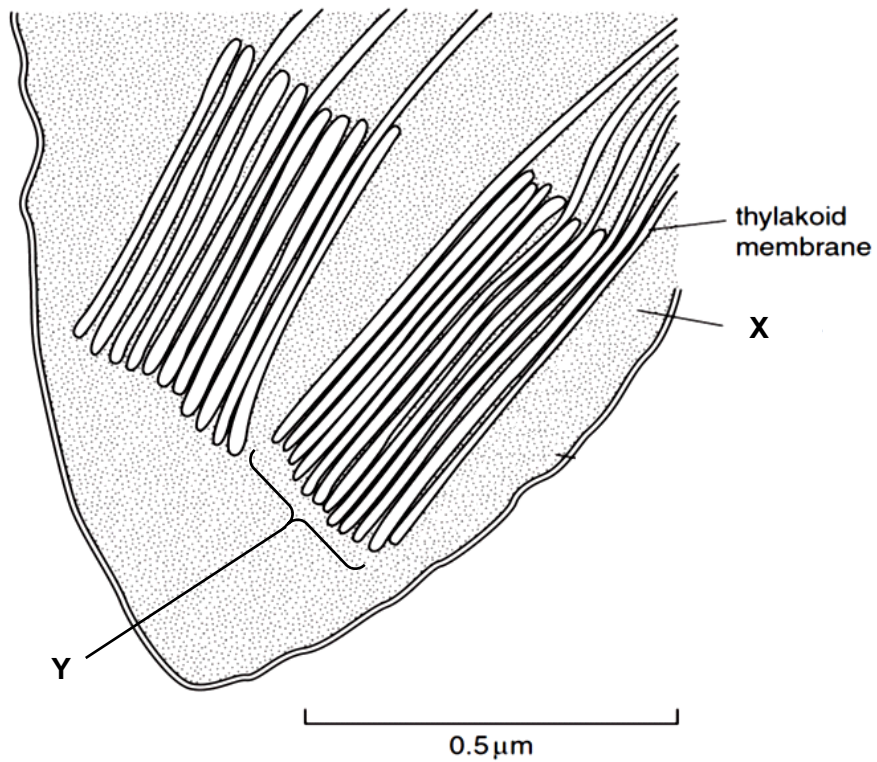


Fig. 6.1

(a) Identify structures X and Y.

X
Y

[2]

Fig. 6.2 shows ATP synthase embedded in the thylakoid membrane.

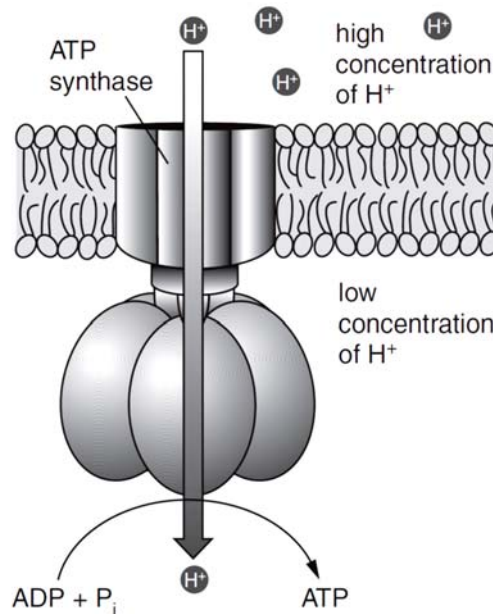


Fig. 6.2

(c) With reference to Fig. 6.3,

- (i) State the total volume of CO₂ used by the plant for photosynthesis when light intensity is no longer the limiting factor.

----- [1]

- (ii) Describe the significance of point **A** to the growth of the plant.

----- [2]

[Total: 11]

- 7 Fig. 7.1 shows the structure of the hormone, insulin, which is responsible for the regulation of blood glucose level back to its normal set point.

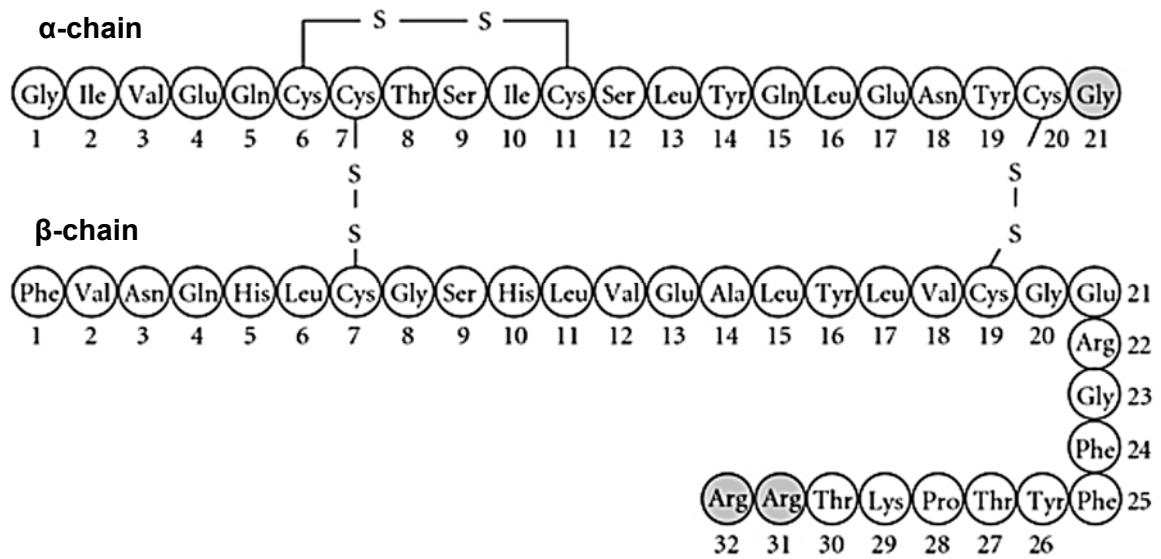


Fig. 7.1

- (a) With reference to Fig. 7.1, describe the structure of insulin.

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.....

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.....

.....

..... [3]

Metformin is used as a drug to treat hyperglycaemic conditions in patients with Type II diabetes where the skeletal muscles are resistant to insulin stimulation. It is transported into the cell by specific protein carrier.

Fig. 7.2 illustrates how metformin influences the cellular activity involved in the signal transduction pathway. AMPK is a kinase that is involved in energy sensing and is activated by AMP (adenosine monophosphate) which is the one of the products of ATP hydrolysis.

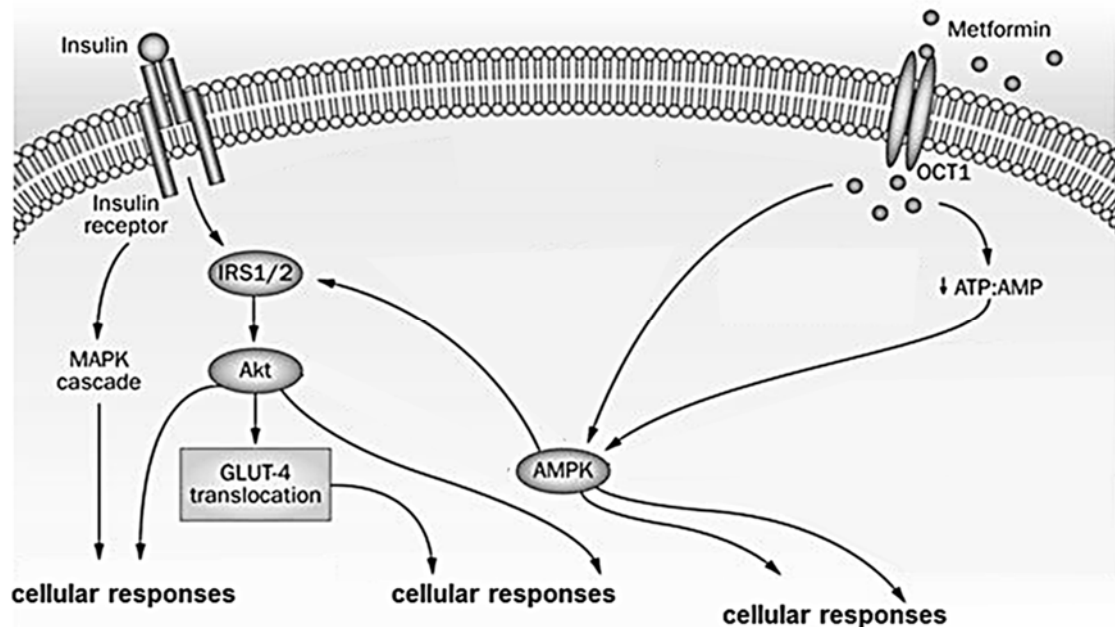


Fig. 7.2

- (b)** With reference to Fig. 7.2,
- (i)** explain how metformin can be used to decrease the blood glucose level in patients with Type II diabetes.

[4]

- (ii) Akt is known to stimulate other cellular responses in the insulin signalling pathway.
Suggest how activation of Akt can lead to different cellular responses.

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.....

.....

..... [3]

[Total: 10]

- 8 A mammoth can be any species belonging to the extinct genus *Mammuthus*, proboscideans commonly equipped with long, curved tusks. They were members of the family Elephantidae which contains, along with mammoths, the two genera of modern elephants and their ancestors.

About three million years ago, the ancestors of mammoths migrated from Africa into Europe and Asia. There, about 1.7 million years ago, the steppe mammoth evolved and became adapted to the cooler conditions. Then, about 700 000 years ago, as the climate changed and the Arctic became much colder, the woolly mammoth evolved.

Woolly mammoths showed a number of obvious adaptations to reduce heat loss, including thick fur, small ears and small tails.

- (a) Explain how variation and natural selection may have brought about the evolution of the woolly mammoth from the steppe mammoth.

[4]

- (b) The phylogenetic relationship of the woolly mammoth, Asian elephant (*Elephas maximus*) and African elephant (*Loxodonta africana*) was then investigated.

Fig. 8.1 shows the phylogenetic tree obtained.

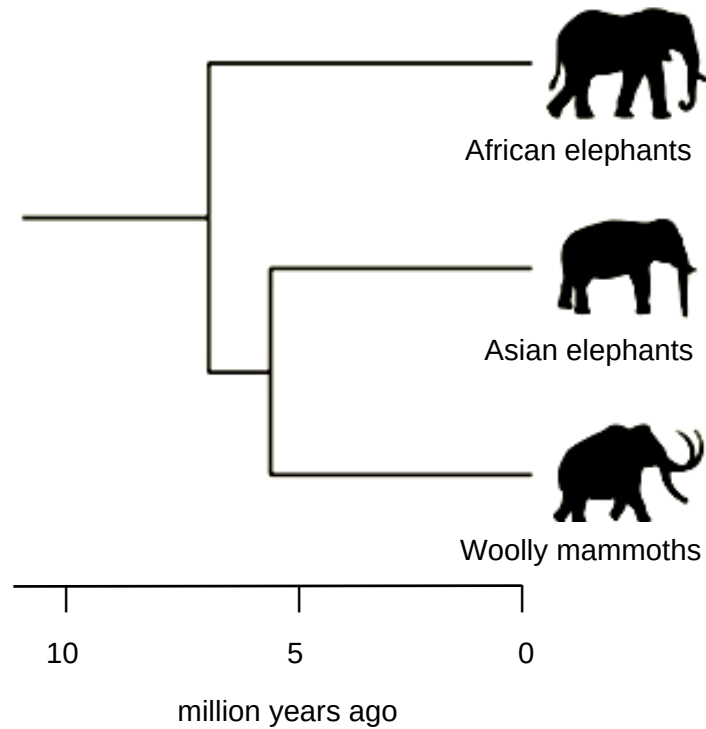


Fig. 8.1

With reference to Fig. 8.1, state

- (ii) when the African elephant diverged;

..... [1]

- (iii) when the Asian elephant diverged.

..... [1]

- (c) Explain how molecular methods can be used to construct the phylogenetic tree in Fig. 10.1.

.....

 [2]

[Total: 8]

- 9 Cholera is an infectious disease caused by the bacterium *Vibrio cholerae*. The bacteria are transmitted between humans through the fecal-oral route. Contaminated food or water can cause infection which symptoms can occur very quickly, from a few hours to a few days.

Table 9 shows

- the economic status of each of five countries
- the number of cases of cholera reported to the World Health Organization (WHO) over a five year period for each country
- the population in 2006 and in 2010 of each country.

Table 9

country	economic status	number of cholera cases reported			population / millions	
		2006	2008	2010	2006	2010
Zimbabwe	low	789	60055	951	12.5297	12.5715
Uganda	low	5194	3726	2341	29.3703	33.4247
Angola	middle	67257	10511	1484	17.0104	19.0819
Cameroon	middle	922	0	10759	17.9484	19.5989
Canada	high	2	1	2	32.6490	34.1088

(a) With reference to Table 9,

- (i) suggest **two** reasons to explain the higher number of cholera cases reported in low economic status countries compared to high economic status country.

----- [2]

- (ii) suggest if economic status of a country is likely to be the main reason for the high number of cases of cholera.

----- [2]

- (b) Cholera can be treated with antibiotics like tetracycline.

- (i) Suggest how tetracycline results in the death of the bacterium.

----- [2]

- (ii) Explain why tetracycline does not cause death of human cells.

----- [1]

[Total: 7]

- 10** Dengue fever is a mosquito-borne viral disease and a regular epidemic in Thailand. The peak of the dengue epidemic period is around June to August during the rainy season. It is believed that climate is an important factor for dengue transmission.

Fig. 10.1 shows how changes in temperature affects the average monthly incidence rate of dengue cases in Chiang Rai, Thailand from 2004 – 2014.

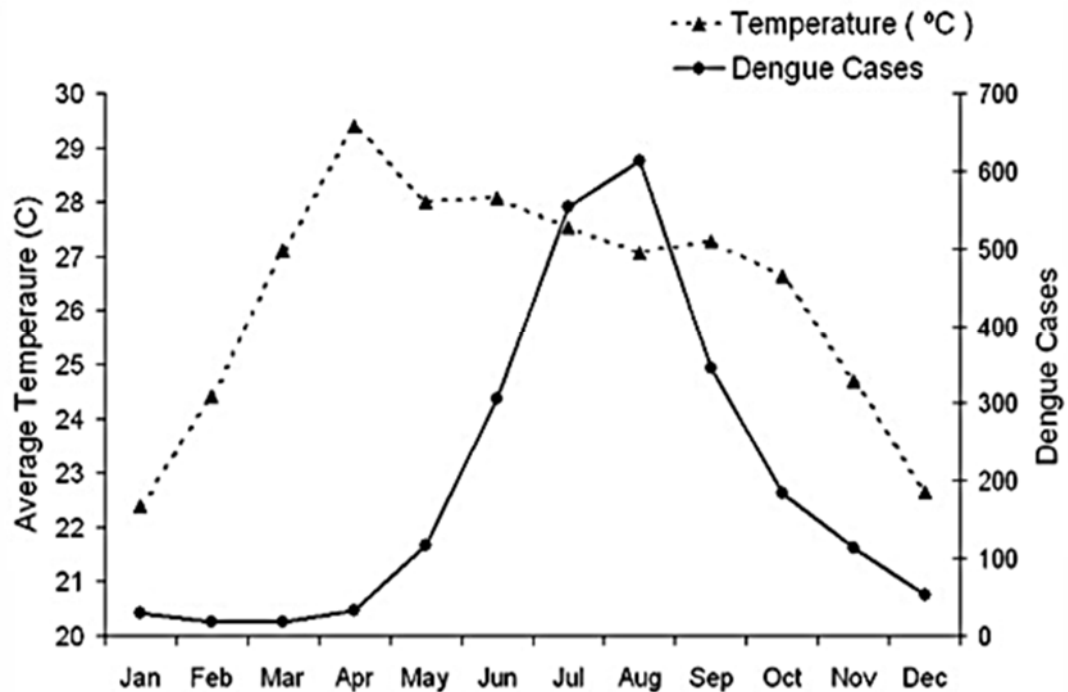


Fig. 10.1

- (a)** State how each data point in Fig. 10.1 is obtained.

[1]

- (b)** With reference to Fig. 10.1,

- (i)** describe the trend of incidence of dengue cases with regard to changes in temperature;

[2]

(ii) explain the trend described in (b)(i).

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.....

.....

.....

.....

..... [3]

[Total: 6]

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